

Practical Assignment-1

Data Types, Variables, and Literals

1. **Program to demonstrate all primitive data types**
Declare variables of `byte`, `short`, `int`, `long`, `float`, `double`, `char`, and `boolean`.
Print their values and memory sizes.
2. **Program to add two integers and display result**
Declare and initialize `int` variables using literals.
3. **Program to calculate area of a circle using `float/double`**
Use literal values like `3.14`.
4. **Program to swap two variables using a third variable (`int`)**

Identifiers and Naming Rules

5. **Program to show valid and invalid identifiers**
Declare variables with proper and improper names (comment out the invalid ones).
Explain camelCase and naming conventions.

Operators

6. **Program to demonstrate all arithmetic operators**
Addition, subtraction, multiplication, division, modulus.
7. **Program to demonstrate relational and logical operators**
Compare two numbers and use `&&`, `||`, `!` with conditions.
8. **Program to demonstrate increment and decrement operators**
Use pre/post-increment and pre/post-decrement.
9. **Program to demonstrate bitwise operators**
Use `&`, `|`, `^`, `~`, `<<`, `>>`.
10. **Program to use compound assignment operators (`+=`, `-=`, etc.)**
Modify a value step-by-step and print intermediate results.

Decision Making & Control Flow Statements

11. **Program to check if a number is even or odd using `if-else`**
12. **Program to find the largest of three numbers using nested `if`**
13. **Program to check if a year is a leap year using `if-else`**
14. **Program to create a simple calculator using `switch-case`**
15. **Program to print day of week using `switch-case`**
 - o Take user input (1–7) and print corresponding day name.
16. **Program to demonstrate the use of `break` and `continue` in a loop**
17. **Program to print numbers from 1 to 10 using `for` loop**
18. **Program to print even numbers from 1 to 100 using `while` loop**
19. **Program to calculate factorial of a number using `do-while` loop**
20. **Program to print multiplication table of any number**